

Dylan Louie

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EDUCATION

University of California San Diego

June 2027

Bachelor of Science in Data Science

Current GPA: 3.95/4.0

Relevant Coursework

Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Probability and Statistics

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML/CSS, SQL

Frameworks: React, Node.js, Express.js

Libraries: Pandas, NumPy, Matplotlib, Scikit-learn, Pillow(PIL), Beautiful Soup

Tools: Git, GitHub, VS Code, REST APIs, OpenAI, Microsoft Office, MATLAB, GitHub Copilot, Claude Code

PROJECTS

League of Legends Match Outcome Predictor | *Python, Scikit-Learn, Pandas*

December 2025

- Analyzed a dataset of 10,000 professional matches to quantify the “snowball effect” of early-game advantages.
- Investigated non-trivial missingness patterns in 118k+ rows of professional LoL match data, utilizing permutation tests to determine dependencies (MAR/NMAR) and conducting Hypothesis Testing to validate the correlation between early-game gold leads and victory.
- Built a Random Forest Classifier to predict match outcomes based solely on 15-minute metrics; implemented GridSearchCV for hyperparameter tuning to resolve overfitting and improved prediction accuracy to 71.1%.
- Developed a robust pipeline including StandardScaler and One-Hot Encoding for league-specific data, while performing a Fairness Analysis to ensure model parity across different game scenarios using absolute accuracy differences.

BeatByBeat | *React, TypeScript, Node.js, Express, Spotify API*

October 2025

- Architected and developed a full-stack, Wordle-inspired music guessing game using React, Node.js, and Express.js with TypeScript for end-to-end type safety
- Engineered a creative “API Mashup” solution by integrating the official Spotify API (for playlist/track data) with the public Deezer API (for audio previews), demonstrating a robust solution to data availability challenges
- Built a dynamic and responsive single-page application (SPA) frontend in React, utilizing hooks (useState, useEffect) and conditional rendering to manage game state and create a seamless, interactive user experience
- Developed a secure and scalable RESTful API with Express.js and Node.js, featuring separate, efficient endpoints for game session creation and on-demand data fetching

NBA Game Outcome Predictor | *Python, Scikit-learn, Pandas, Matplotlib, Jupyter Notebook*

July 2025

- Engineered an end-to-end machine learning pipeline to predict NBA game winners, achieving 63.2% accuracy on a held-out test set from the 2022-2023 season.
- Automated the web scraping of 17,000+ games from Basketball Reference using a Python pipeline with Playwright and BeautifulSoup, then leveraged Pandas to clean, transform, and engineer over 20 predictive features.
- Trained and evaluated multiple classification models (e.g., Logistic Regression, Random Forest), using Scikit-learn to compare performance and select the optimal algorithm.
- Generated data visualizations using Matplotlib to perform feature importance analysis, identifying key statistical factors that influence game outcomes.

Python Image & Audio Manipulation Engine | *Python, NumPy, Pillow (PIL)*

March 2025

- Developed a comprehensive Python toolkit utilizing object-oriented principles to perform complex manipulations on image and audio data.
- Implemented a suite of image processing filters from scratch, including grayscale, rotation, and an edge-highlight function using a convolution kernel.
- Built a K-Nearest Neighbors (KNN) image classifier from the ground up, engineering the Euclidean distance and voting mechanisms to classify images based on pixel data.
- Extended the toolkit to process WAV audio files, adding features for reversing, time-stretching (speed up/slow down), and applying a reverb effect.